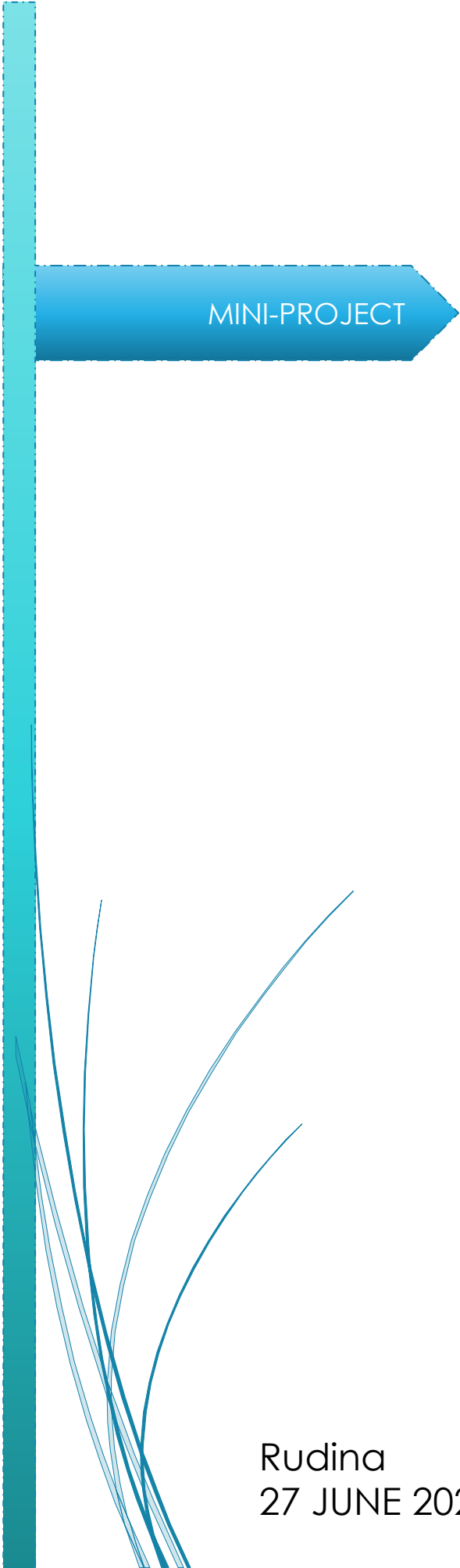
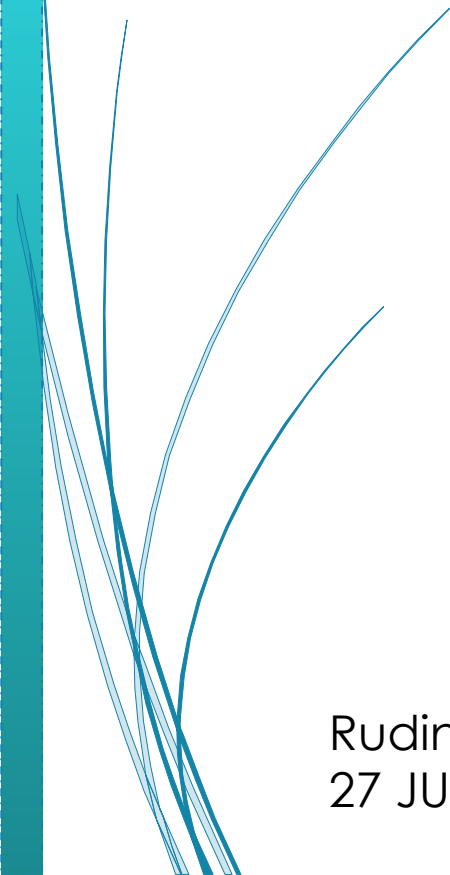


A thick vertical bar on the left side of the page, transitioning from light blue at the top to dark teal at the bottom. A blue arrow points from this bar towards the right, containing the text 'MINI-PROJECT'.

MINI-PROJECT

Customer Segmentation Model

ANALYSIS REPORT

Several thin, curved, light blue lines that originate from the bottom left and sweep upwards and to the right, resembling stylized grass or reeds.

Rudina
27 JUNE 2026

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OVERVIEW

1. Introduction

This project applies Machine Learning (ML) to support an e-commerce company’s marketing decisions by developing a customer segmentation model based on customer behaviour, value and engagement.

K-Means clustering is applied to a transnational dataset covering customers from five continents and 47 countries, providing insights into purchasing behaviour and preferences across a diverse international customer base.

2. Methods

This project segments customers by purchase behaviour using data preparation, duplicate removal, feature engineering, EDA, IQR outlier detection, log transformation, scaling, cluster-number evaluation, K-Means clustering and PCA/T-SNE visualisation (see Table 1). The 1-5% outlier requirement was used as the stopping criterion for anomaly detection.

Table 1: Summary of techniques used during this project.

TECHNIQUE	TYPE	DESCRIPTION	PARAMETERS USED
IQR	Statistical method	Detects outliers in skewed data.	5 new features
Elbow method	Learning algorithm	Tests cluster numbers using within-cluster variation.	Inertia/WCSS
Silhouette method	Quantitative metric	Assesses cluster cohesion and separation.	Silhouette score
Hierarchical clustering/Dendrogram	ML	Shows cluster relationships as a tree.	Ward linkage
Davies-Bouldin index (DBI)	Quantitative metric	Measures cluster compactness and separation.	Davies-Bouldin score
K-Means Clustering Model	ML	Groups customers by feature similarity.	Number of clusters (k)
PCA	Dimensionality reduction (DR) techniques	Linear DR method using maximum variance components.	1 st and 2 nd PC.
T-SNE		Non-linear DR method for visualising cluster structure.	Perplexity with values (5 – 35) (Xiao et al., 2023).

Figure 1 illustrates the overall analytical workflow for this project with each stage labelled according to the method applied during the segmentation process.

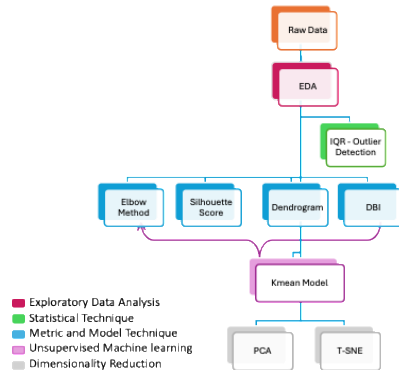


Figure 1: Diagram of various approaches used during this project for customer segmentation purpose.

3. Results and Discussion

During this analysis, 21 fully duplicate rows were identified and removed. After removal, the cleaned dataset contained 951,648 rows across 20 columns, representing 68,300 unique customers.

Table 2: Summary statistics for 5 newly engineered features in the dataset.

	Frequency	CLV	Avg_Unit_Cost	Age	Recency
count	68300.0	68300.0	68300.0	68300.0	68300.0
mean	13.9	1950.1	78.9	41.2	367.7
std	11.3	1719.9	38.0	17.5	414.5
min	1.0	0.6	0.5	14.0	0.0
25%	6.0	696.0	57.6	25.3	45.0
50%	11.0	1497.4	73.5	40.8	190.0
75%	19.0	2709.8	92.5	56.2	539.3
max	121.0	18860.96	1463.5	74.0	1824.0

Following EDA, five engineered features were grouped into a new dataset (Table 2). The statistics show clear customer heterogeneity, while Figure 2 shows that four features are right-skewed with upper outliers, reflecting typical retail behaviour where a small number of high-value customers drive the right tail.

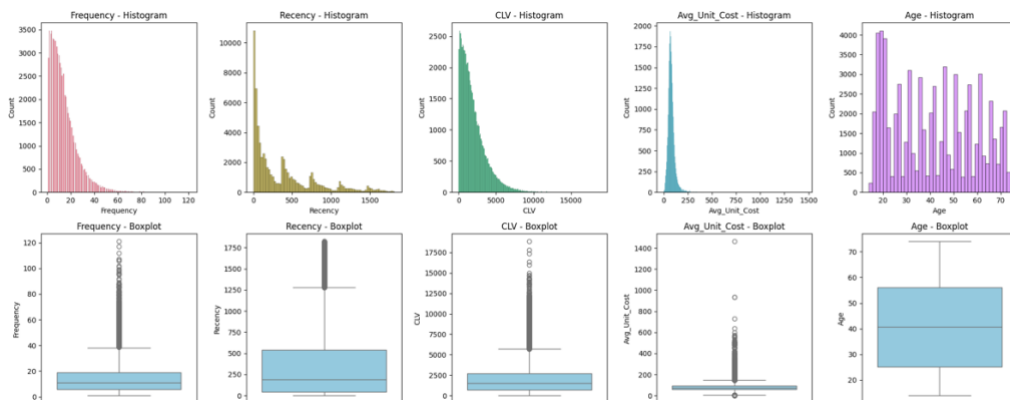


Figure 2: New 5 Feature distribution plots and boxplots reveal the skewed distribution patterns and the outliers.

3.1 IQR

During IQR outlier detection it was observed that the outlier's percentage for each feature was below 4% with Recency close to 5%, see Table 3.

Feature	Q1	Q3	Lower Bound	Upper Bound	Outliers	Outlier %
Frequency	6.00	19.00	-13.50	38.50	2654	3.9
Recency	45.00	539.25	-696.38	1280.62	3361	4.9
CLV	696.00	2709.84	-2324.77	5730.61	2590	3.8
Avg_Unit_Cost	57.58	92.52	5.18	144.92	2889	4.2
Age	25.25	56.19	-21.15	102.59	0	0.0

Table 3: IQR values falling below lower bound and above upper bound were calculated and number of outliers are shown as a percentage.

These outliers were not removed as extreme values often represent genuinely high-value or highly engaged customers.

3.2 KMeans Model Evaluation

K-Means model was evaluated using several validation methods. Figure 4A shows no clear elbow, so the optimal value of k was not obvious from within-cluster variation.

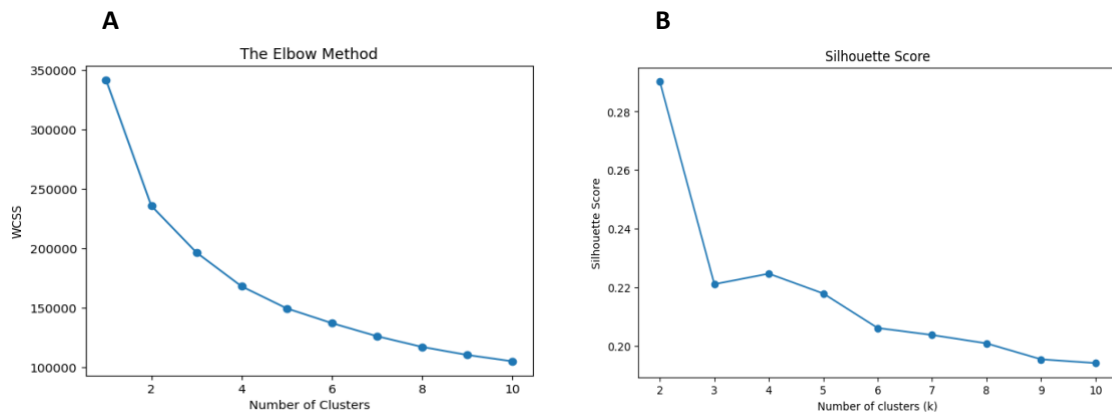


Figure 4: A) Elbow method shows that number of clusters shows not clearly defined elbow shape. However, B) Silhouette score indicated that the best option could be $k = 5$.

Silhouette coefficient of 0.225 suggested that $k = 4$ was a strong candidate because it produced well-separated and cohesive clusters. However, the DBI as shown in Figure 6, reached its lowest value 1.289 at $k = 5$, confirming reasonable cluster compactness according to this metric.

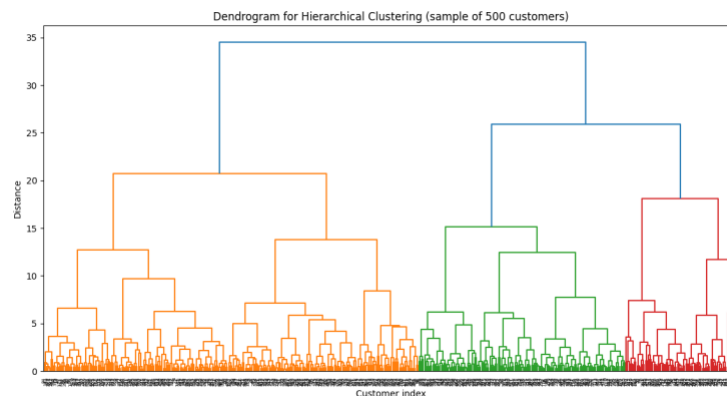


Figure 5: Visual representation through a dendrogram of possible cluster creation within the data.

Similar with the dendrogram in Figure 5, several major branches separating at relatively large distances at $k = 5$. Although $k = 4$ may be statistically defensible, $k = 5$ was selected for the final K-Means model because it provided a more detailed and interpretable segmentation structure for marketing decision-making. This approach balances statistical evidence with interpretability for business relevance.

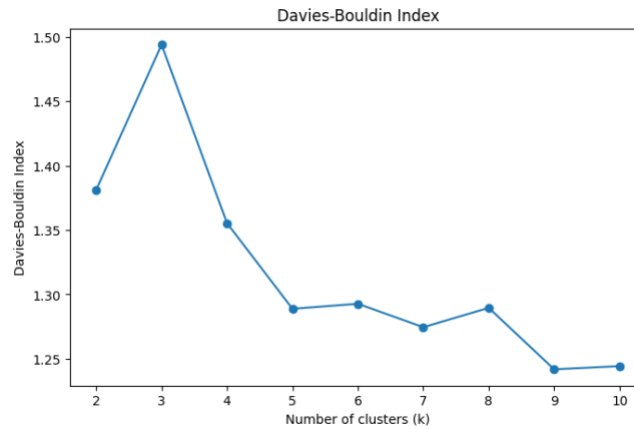


Figure 6: Evaluation of optimum number of k using DBI Score showing k number of clusters.

4. DR

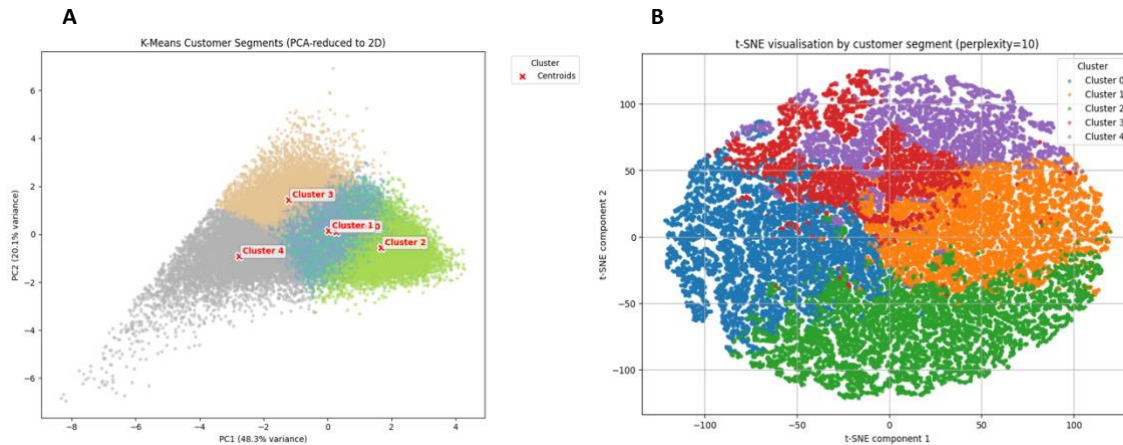


Figure 7: Panel A and B show PCA and T-SNE visualisation of the five-segment customer segmentation.

Both PCA and T-SNE visualisation reveal that customers do not fall into sharply discrete clusters rather than show overlapping cluster boundaries.

5. Conclusion

In conclusion, this mini-project involved pre-processing the real-life raw data, engineering customer-level features, conducting EDA analysis before applying K-Means customer segmentation model. Four clustering evaluation methods were used to identify the optimal number of clusters for log scaled dataset. Although the Silhouette result suggested that four clusters may be statistically strong, the dendrogram and DBI provided support for selecting $k = 5$ as the final solution.

Dimensionality reduction using PCA and T-SNE showed that the five customer segments were not clearly separated in 2D space. T-SNE provided better visual separation than PCA, where PC1 explained 48% of variance and PC2 explained 20.1%. However, the overlapping cluster structure can be partially explained by the presence of outliers and the strong positive

collinearity between Frequency and CLV. For future work, we could therefore test feature selection or further dimension reduction approaches to improve cluster separation and model interpretability.

7. Reference

1. SAS, 2024. CUSTOMERS_CLEAN [Data set]. SAS. Last revised on 15 December 2021. [Accessed 20 February 2024].
2. Influence of Data Dimension Reduction, Feature Scaling and Activation Function on Machine Learning Performance. (Słowiński, et al., 2021)
3. Optimizing graph layout by t-SNE perplexity estimation. (Xiao et al., 2023).